Unbundling LLM Recommenders: Retrieval, Persuasion, and Consumer Choice

Qifan Han*

This draft: May 12, 2026

Abstract

LLM-mediated shopping assistants bundle two economically distinct functions: they select which product to recommend and they generate the language that justifies the recommendation. A standard backend prompt A/B test identifies the total effect of a policy change, but it generally does not reveal whether the effect comes from product retrieval, from persuasive expression, or from the matching between the two. In this paper, we propose a modular 2×2 experimental design that crosses a retrieval policy $Q \in \{0, 1\}$ with an expression policy $R \in \{0, 1\}$, and we show that cell-mean contrasts on the design identify the retrieval effect, the persuasion effect, and their interaction over a feasible recommendation support. We then use a controlled structural simulation calibrated to three product categories to demonstrate two findings. First, we find that the modular design recovers the three components in two qualitatively different data-generating processes, while standard one-shot prompt A/B tests deliver only the total effect. Second, we find that a naive regression of outcomes on realized expression intensity overstates the persuasion effect by 80–130% of the truth across the policy-relevant range of expression endogeneity, because realized expression is endogenously correlated with latent product fit. Taken together, the design clarifies which lever a firm has actually pulled when an LLM recommendation policy moves consumer choice.

Keywords: LLM recommendation, factorial design, retrieval, persuasion, latent textual treatments.

1 Introduction

LLM shopping assistants are becoming the front door of product discovery. A consumer can ask a conversational agent for a laptop, a coffee maker, or a travel-insurance plan and receive a natural-language recommendation in seconds.[1] Unlike a conventional ranker, the assistant does not merely retrieve a list of products. It recommends one or more products and generates the language that explains why the recommendation fits the consumer.

In this paper, we study how firms can decompose the causal effect of an LLM recommendation policy into a *retrieval* component and a *persuasion* component. A backend policy change at a retailer's LLM raises the click rate. The retailer needs to know whether the lift came from selecting better-fitting products, from writing more persuasive explanations, or from matching particular

*Department of Marketing.

[1] A LLM-mediated shopping assistant, throughout this paper, is any chatbot interface that returns one or more product suggestions accompanied by generated explanatory text. The category includes ChatGPT's shopping experience, Perplexity, Amazon's Rufus, and many smaller retailer-deployed agents.

expressions to particular products. These correspond to different investments: better catalog grounding, better prompt engineering, or product-specific expression policies.

On the one hand, the standard randomization argument does most of what a firm wants. Randomizing the backend prompt across consumers identifies the total effect of the policy change. On the other hand, the same total effect can be generated by very different mixes of retrieval-channel and expression-channel lifts, and these mixes correspond to different investment decisions. We therefore frame the central question as: *when an LLM recommender changes consumer choice, is it acting as a retrieval engine, a persuasion engine, or both at once?*

Our framework is built around a modular 2×2 experimental design. The first stage selects a product under a *retrieval policy* Q. The second stage generates recommendation text under an *expression policy* R conditional on the selected product. Crossing Q and R creates four cells, including the hybrid regimes $(Q = 1, R = 0)$ and $(Q = 0, R = 1)$ that a one-shot policy does not realize. We show that the cell-mean contrasts of the four cells identify the retrieval effect Δ_R, the persuasion effect Δ_P, and their interaction $\Delta_{R\times P}$, while a one-shot prompt A/B test generally identifies only the total effect.

We assemble a controlled structural simulation built on three hand-curated product catalogs (phone chargers, headphones, and laptops) and 1,000 simulated consumers per category, each with a latent product-consumer fit score Q_{ij}. We simulate one-shot and modular recommendations and outcomes under two qualitatively different data-generating processes, one in which persuasion is the leading channel and one in which retrieval is. We then estimate the modular cell-mean contrasts and three competing estimators of the persuasion effect: a naive regression of outcomes on realized expression intensity, an infeasible oracle regression that additionally controls for the latent fit score, and the modular cell contrast.

First, we find that the modular design recovers the retrieval, persuasion, and interaction components within sampling noise in both data-generating processes. In the persuasion-leaning DGP, the pooled persuasion effect is estimated at 0.032 (true value 0.037, $t = 2.56$); the retrieval effect washes out by construction. In the retrieval-dominant DGP, the pooled retrieval effect is estimated at 0.018 (true value 0.014, $t = 1.44$) and the persuasion effect is estimated at 0.029 (true value 0.023, $t = 2.26$).

Second, we find that the naive realized-expression regression is materially biased. The bias is non-monotonic in λ_{fit}, the strength of the endogenous correspondence between latent fit and generated expression: it rises sharply over $\lambda_{\text{fit}} \in [0, 0.75]$ to a peak of approximately 0.05, then attenuates at higher λ_{fit}. At policy-relevant endogeneity levels the naive estimator overstates the persuasion effect by 80–130% of the truth. The oracle regression and the modular cell contrast are unbiased throughout. We interpret the non-monotonic shape as the classical omitted-variable bias pattern, in which the bias is proportional to $\text{Cov}(E, Q)/\text{Var}(E)$ — a ratio whose numerator is linear in λ_{fit} but whose denominator is quadratic.

To the best of our knowledge, this paper is the first to formalize the retrieval-versus-persuasion decomposition for LLM-mediated product recommendation as an experimentally implementable factorial design. Our findings provide at least threefold insights for marketing research and practice. First, standard prompt A/B tests answer the wrong question when the decomposition matters: they identify the total effect of a policy change, but they leave the underlying lever bundled. Second, a researcher who regresses outcomes on realized expression intensity is, under any positive expression-fit endogeneity, working with a biased estimator. Third, the modular design is deployable at the platform layer with only minor changes to the backend pipeline: it requires separable retrieval and expression calls and an experimental allocation that is uncorrelated with the consumer context. Taken together, these findings suggest that firms can move from "the LLM helped" to "this layer of the LLM helped" with a small, well-structured platform experiment.

The remainder of the paper is structured as follows. Section 2 positions the paper in the literature on causal inference with text and on platform experimentation. Section 3 introduces the setting and the notation. Section 4 states and discusses the four identification results. Section 5 describes the modular 2×2 experiment. Section 6 presents the simulation evidence. Section 7 concludes.

2 Related Literature

This paper contributes to two strands of existing literature.

The first studies the causal interpretation of text and generative content. Egami et al. [2018] formalize how textual data can be used as a treatment or as an outcome in causal designs. Feldman et al. [2026] develop an estimator for the causal effect of latent textual features, and show that naive regressions on realized text can be biased because realized text conflates the treatment with covariate information. Nakamura and Imai [2025] propose a generative-probability-of-intervention framework that uses LLM outputs as treatments in observational studies. Reisenbichler et al. [2022] use natural language generation as a marketing-content production technology. Angelopoulos et al. [2024] align language-model outputs to causal signals from A/B tests in order to improve marketing communications. We extend this literature by studying a recommendation-specific decomposition: LLM-mediated product recommendations bundle a discrete product-selection decision with a continuous expression decision, and a modular experimental design can identify each as a local policy effect.

The second strand studies conversational recommendation, LLM adoption, and platform experimentation. Deldjoo [2024] documents that prompt design, system role, and intent affect ChatGPT-based top-K recommendation accuracy, provider fairness, and temporal stability. Jannach [2023] surveys evaluation conventions for conversational recommender systems. Johnson et al. [2017] introduce ghost ads as a way to identify the counterfactual exposure of digital ads, an idea we borrow from in spirit when restricting modular effects to a feasible recommendation support. Berman [2018] and Li and Kannan [2014] address the related problem of attributing conversions across multiple advertising touchpoints. We contribute to this literature by treating retrieval and persuasion as two separately manipulable policy components of a single recommendation event, and by providing a deployable factorial experiment that estimates each as a local policy effect.

3 Setup and Notation

3.1 Setting

A consumer-session indexed by i arrives at an LLM-mediated shopping interface, submits a query, and receives a natural-language recommendation. The interface returns a tuple (J_i, E_i) — a selected product $J_i \in \mathcal{A}_i$ and an expression state E_i — and the consumer takes an action $Y_i \in \{0, 1\}$ (click, purchase, or search-continuation).[2] A backend policy controls both layers of the interface. We treat each consumer-session as an independent unit, and our object of interest is the average causal effect of changing the backend policy.

3.2 Random variables and potential outcomes

We adopt the potential-outcomes framework. For each consumer-session i:

[2] In the simulation we use binary outcomes for tractability; the framework applies to any scalar outcome.

- X_i denotes the pre-treatment context (the query, the conversation history before the recommendation, the consumer's stated preferences, the budget, and any session covariates available before recommendation);
- $\mathcal{A}_i \subseteq \mathcal{J}$ denotes the feasible product catalog for the session;
- $J_i \in \mathcal{A}_i$ is the recommended product;
- $E_i \in \mathbb{R}$ is the realized expression intensity;[3]
- $Y_i \in \{0, 1\}$ is the outcome.

We write the joint potential outcome as $Y_i(j, e)$, the outcome that consumer-session i would realize if shown product j with expression state e. Policy assignments are denoted $Z_i \in \{0, 1\}$ for the one-shot bundled policy and $(Q_i, R_i) \in \{0, 1\}^2$ for the modular policies. Realized assignments determine which potential outcome is observed under SUTVA and consistency.

3.3 Policy kernels

A backend policy induces a distribution over (j, e) pairs through a stochastic kernel. We write the one-shot kernel under policy z as

$$K_z^{1\mathrm{s}}(j, e \mid x, \mathcal{A}) = \Pr(J_i = j, E_i = e \mid X_i = x, \mathcal{A}_i = \mathcal{A}, Z_i = z). \tag{1}$$

By the chain rule, this kernel factorizes as $K_z^{1\mathrm{s}}(j, e \mid x, \mathcal{A}) = K_z^{1\mathrm{s}J}(j \mid x, \mathcal{A}) \cdot K_z^{1\mathrm{s}E}(e \mid x, \mathcal{A}, j)$.

The modular kernel is the same product, but with two independent policy indices:

$$K_{q,r}^{2\mathrm{s}}(j, e \mid x, \mathcal{A}) = K_q^J(j \mid x, \mathcal{A}) \cdot K_r^E(e \mid x, \mathcal{A}, j). \tag{2}$$

The substantive difference between Equation (1) and Equation (2) is the bundle structure: in the one-shot kernel, z enters both the retrieval factor and the expression factor; in the modular kernel, q enters only the retrieval factor and r enters only the expression factor. The asymmetry is what allows the modular design to identify quantities that the one-shot design cannot.

4 Identification

This section states the four identification results that anchor the paper. Proofs are given in Appendix A. Throughout, we maintain SUTVA and consistency, randomized policy assignment, and overlap; we state any additional condition explicitly inside each Proposition.

4.1 Estimands

The four estimands of interest are defined on the modular kernel:

$$\Delta_{\text{total}} = \mathbb{E}\Big[\textstyle\sum_{j,e} K_1^{1\mathrm{s}}(j, e \mid X, \mathcal{A}) Y(j, e)\Big] - \mathbb{E}\Big[\textstyle\sum_{j,e} K_0^{1\mathrm{s}}(j, e \mid X, \mathcal{A}) Y(j, e)\Big] \tag{3}$$

$$\Delta_R = \mathbb{E}\Big[\textstyle\sum_{j,e} K_{1,0}^{2\mathrm{s}}(j, e \mid X, \mathcal{A}) Y(j, e)\Big] - \mathbb{E}\Big[\textstyle\sum_{j,e} K_{0,0}^{2\mathrm{s}}(j, e \mid X, \mathcal{A}) Y(j, e)\Big] \tag{4}$$

$$\Delta_P = \mathbb{E}\Big[\textstyle\sum_{j,e} K_{0,1}^{2\mathrm{s}}(j, e \mid X, \mathcal{A}) Y(j, e)\Big] - \mathbb{E}\Big[\textstyle\sum_{j,e} K_{0,0}^{2\mathrm{s}}(j, e \mid X, \mathcal{A}) Y(j, e)\Big] \tag{5}$$

$$\Delta_{R\times P} = \mathbb{E}\big[Y(K_{1,1}^{2\mathrm{s}}) - Y(K_{1,0}^{2\mathrm{s}}) - Y(K_{0,1}^{2\mathrm{s}}) + Y(K_{0,0}^{2\mathrm{s}})\big] \tag{6}$$

[3] We model E as a real-valued scalar throughout the analytical results in this paper. The intensity is meant to encode such observable features as endorsement strength, comparative framing, or quality emphasis. Section 6 uses a scalar specification; the framework extends to multivariate E without altering the identification results.

4.2 Identification of the total one-shot effect

Proposition 1 (Prompt randomization identifies the total one-shot effect)**.** *Under SUTVA, consistency, and randomized assignment of the one-shot policy Z, the total one-shot effect Δ_{total} defined in (3) is identified by*

$$\Delta_{\text{total}} = \mathbb{E}[Y \mid Z = 1] - \mathbb{E}[Y \mid Z = 0],$$

possibly after conditioning or reweighting over X and $\mathcal{A}$.

Interpretation. A standard prompt A/B test identifies the total policy effect on average. We treat this result as a benchmark, not a contribution: it is the baseline from which the rest of the paper departs.

4.3 One-shot is not a decomposition

Proposition 2 (One-shot non-identification of components)**.** *Under the conditions of Proposition 1, the difference $\mathbb{E}[Y \mid Z = 1] - \mathbb{E}[Y \mid Z = 0]$*

$$\Delta_{\text{total}} \overset{\textit{generally}}{\neq} \Delta_R + \Delta_P + \Delta_{R\times P},$$

and, more strongly, it does not identify Δ_R, Δ_P, or $\Delta_{R\times P}$ separately.

Interpretation. The proof in Appendix A.2 exhibits two distinct triples $(\Delta_R, \Delta_P, \Delta_{R\times P})$ that produce the same value of Δ_{total}. The implication is that a researcher who reports the prompt A/B coefficient is silent on the underlying lever, even when the coefficient is precisely estimated.

4.4 Modular identification

Proposition 3 (Modular factorial identification)**.** *Under SUTVA, consistency, randomized assignment of (Q, R) over the feasible support $\mathcal{A}_i$, and overlap (each modular cell has positive probability for the relevant context), the modular cell-mean contrasts identify the retrieval, persuasion, and interaction effects:*

$$\begin{aligned}
\Delta_R &= \mathbb{E}[Y \mid Q = 1, R = 0] - \mathbb{E}[Y \mid Q = 0, R = 0],\\
\Delta_P &= \mathbb{E}[Y \mid Q = 0, R = 1] - \mathbb{E}[Y \mid Q = 0, R = 0],\\
\Delta_{R\times P} &= \mathbb{E}[Y \mid Q = 1, R = 1] - \mathbb{E}[Y \mid Q = 1, R = 0]\\
&\quad - \mathbb{E}[Y \mid Q = 0, R = 1] + \mathbb{E}[Y \mid Q = 0, R = 0].
\end{aligned}$$

Interpretation. The factorial design identifies the components because the four cells span the four corners of $\{(Q, R)\}$, and randomized assignment makes each cell mean estimate the expected potential outcome under the corresponding modular policy. The identification is local to the experimental support: products and consumer contexts for which the modular kernel has zero or near-zero probability are not part of the estimand.

4.5 Naive realized-expression regressions are biased

Proposition 4 (Naive realized-expression bias)**.** *Suppose E_i is generated by a kernel of the form $E_i = e_0 + \tau R_i + \lambda_{\text{fit}}\tilde{Q}_{i,J_i} + \sigma_E \eta_i$, where $\tilde{Q}_{i,J_i}$ is a latent product-consumer fit channel that affects the*

outcome through $Y_i = \beta_0 + \beta_E E_i + \beta_Q \tilde{Q}_{i,J_i} + \nu_i$*, and the researcher does not observe* $\tilde{Q}_{i,J_i}$*. Then the OLS coefficient of* Y_i *on* E_i *converges in probability to*

$$\beta_E^{\text{naive}} \;=\; \beta_E + \frac{\text{Cov}(E, \tilde{Q})}{\text{Var}(E)}\beta_Q \;=\; \beta_E + \frac{\lambda_{\text{fit}}\text{Var}(\tilde{Q})}{\lambda_{\text{fit}}^2\text{Var}(\tilde{Q}) + \sigma_E^2}\beta_Q,$$

and the bias is positive whenever $\lambda_{\text{fit}}\beta_Q > 0$*.*

Interpretation. The bias has a closed form, and the closed form is non-monotonic in λ_{fit}. The numerator of the bias term is linear in λ_{fit}, but the denominator is quadratic. The bias therefore rises to a maximum at $\lambda_{\text{fit}}^\star = \sigma_E/\sqrt{\text{Var}(\tilde{Q})}$ and falls back toward zero at the extremes of λ_{fit}. Section 6 presents Figure 1 which traces this curve empirically. The pattern is sometimes mistaken for a numerical artifact; we read it as a feature of the bias formula, and we return to it in the Conclusion.

5 Proposed Framework: A Modular 2×2 Experiment

The framework operationalizes Proposition 3 as a deployable platform experiment.

5.1 Two-stage architecture

The experiment has two stages.

Stage 1 (retrieval). Take the consumer context X_i and the feasible catalog $\mathcal{A}_i$. Apply retrieval policy $Q_i \in \{0, 1\}$. Return a product $J_i \in \mathcal{A}_i$. In a deployed system, this stage corresponds to a JSON-mode call that returns a product identifier and a shortlist, without persuasive recommendation text.

Stage 2 (expression). Take the product J_i returned by Stage 1 along with the consumer context. Apply expression policy $R_i \in \{0, 1\}$. Return the recommendation text. In a deployed system, this stage corresponds to a free-form generation call that may not change the selected product.

The architecture has three deployment requirements. First, the platform must be able to issue separable retrieval and expression calls, which is now standard. Second, the expression stage cannot rewrite the product selected in Stage 1; in practice, this is enforced through the prompt and verified ex post. Third, the modular cell assignment must be uncorrelated with the consumer covariates, which is satisfied by experimenter-controlled randomization.

5.2 Estimands and managerial mapping

Three estimands, one per managerial decision.

The *retrieval effect* Δ_R is the probability-scale change in outcomes when the firm switches from a baseline retrieval rule to an alternative retrieval rule, holding the expression rule fixed. It maps to an investment in catalog grounding, retrieval-side fine-tuning, or improved consumer profiling.

The *persuasion effect* Δ_P is the probability-scale change in outcomes when the firm switches from a baseline expression rule to an alternative expression rule, holding the retrieval rule fixed. It maps to an investment in prompt engineering, copywriting templates, or expression-side fine-tuning.

The *retrieval-persuasion interaction* $\Delta_{R\times P}$ is the difference of the persuasion effect across retrieval rules. It maps to an investment in matched product-expression strategies: e.g., persuasive language for entrant brands but neutral language for incumbents. The sign of $\Delta_{R\times P}$ tests whether persuasive expression substitutes for or complements brand equity in the studied environment.

The design affects experimental allocation, not identification. The platform retains complete control over the products in the catalog, the prompts used in each stage, and the consumer routing.

6 Simulation Evidence

We present structural simulation evidence in two passes. Pass 1 (Section 6.2) re-derives the prompt-A/B total effect as a benchmark and shows that, while unbiased on average, it is silent on the decomposition. Pass 2 (Section 6.3) presents the modular cell-mean contrasts on two data-generating processes — one in which persuasion is the leading channel and one in which retrieval is. Section 6.4 presents the central bias result: the naive realized-expression regression versus the modular cell contrast across a range of expression-fit endogeneity. Section 6.5 previews the LLM-based simulation and is left as a placeholder for the next round.

6.1 Simulation setup

We construct three product categories of varying complexity: *phone chargers* (8 products, low complexity), *headphones* (10 products, medium complexity), and *laptops* (10 products, high complexity). Each catalog is hand-curated with brand, price, quality, use-case fit, and an incumbent-versus-entrant indicator. We generate 1,000 consumer profiles per category with lognormal budgets, uniform price and quality sensitivities, and Beta-distributed brand familiarity. For each consumer-product pair we compute a latent fit score $Q_{ij} \in [0, 1]$ that we treat as ground truth throughout the simulation.[4]

We simulate recommendation, expression, and outcomes under two data-generating processes. The *persuasion-leaning* DGP uses a brand-forward alternative retrieval policy that substitutes brand affinity for fit; the persuasion channel is the leading channel. The *retrieval-dominant* DGP uses a fit-aware alternative retrieval policy that raises the Q-weighting under $Q = 1$; both retrieval and persuasion are positive, with retrieval as the leading channel. We hold the master seed and the consumer and catalog data fixed across DGPs.[5]

6.2 Pass 1: total-effect prompt A/B

Table 1 reports the total one-shot effect Δ_{total} in the persuasion-leaning DGP.

Table 1: Total one-shot effect, persuasion-leaning DGP.

Category	True Δ_{total}	Estimate	SE	t-stat
headphones	0.032	0.000	0.022	0.00
laptop	0.023	0.028	0.021	1.31
phone_charger	0.034	−0.003	0.021	−0.14
POOLED	**0.030**	**0.008**	**0.012**	**0.67**

The pooled estimate is unbiased in expectation but underpowered at this sample size. The point of Pass 1 is not that the prompt A/B fails to identify the total effect — it does identify it on average, as Proposition 1 states. The point is that even when the prompt A/B is well-powered, the resulting coefficient is silent on the decomposition. We turn next to that decomposition.

[4] Catalog and consumer-generation procedures are reproduced in Appendix C. The latent fit score is a weighted sum of price-fit, quality-fit, use-case fit, and brand familiarity; the parameters are documented in Appendix C.

[5] Full DGP equations and parameter values are reproduced in Appendix B. Seed handling is described in Appendix B.4. The simulation produces $N = 6{,}000$ one-shot observations and $N = 12{,}000$ modular observations per DGP, pooled across the three categories.

6.3 Pass 2: modular decomposition

Persuasion-leaning DGP. Pooled across the three categories, the modular cell contrasts recover the persuasion effect at $t = 2.56$ (estimate 0.032 against truth 0.037). The retrieval effect is near zero in both truth and estimate, by construction: the brand-forward alternative substitutes brand affinity for fit, so it shifts product shares (TVD between $Q = 0$ and $Q = 1$ shares is 0.23–0.31 by category) without systematically improving fit. The interaction is small and statistically indistinguishable from zero.

Table 2: Modular decomposition, pooled across categories.

	Persuasion-leaning DGP			Retrieval-dominant DGP		
Component	True	Estimate (t)	Bias	True	Estimate (t)	Bias
Retrieval Δ_R	-0.006	-0.004 (-0.29)	$+0.002$	$+0.014$	$+0.018$ ($+1.44$)	$+0.004$
Persuasion Δ_P	$+0.037$	$+0.032$ ($+2.56$)	-0.005	$+0.023$	$+0.029$ ($+2.26$)	$+0.006$
Interaction $\Delta_{R\times P}$	-0.000	$+0.018$ ($+1.04$)	$+0.019$	-0.000	-0.006 (-0.32)	-0.006

Retrieval-dominant DGP. In the second DGP the fit-aware alternative retrieval policy genuinely improves consumer-product fit. The retrieval effect is now positive in both truth ($+0.014$) and estimate ($+0.018$, $t = 1.44$). The persuasion effect remains positive ($t = 2.26$). The interaction is again near zero.[6]

6.4 The naive estimator overstates persuasion

Table 3 reports three competing estimators of the persuasion effect, on a comparable probability scale (the effect of moving $R = 0 \to R = 1$ at $Q = 0$).

Table 3: Naive vs. oracle vs. modular estimators of Δ_P (pooled, $\lambda_{\text{fit}} = 1.0$).

Estimator	Scaled estimate	True Δ_P	Bias
Naive realized-E regression	0.087	0.037	**+0.050**
Oracle (adds $\tilde{Q}_J$)	0.040	0.037	$+0.003$
Modular 2×2 cell contrast	0.032	0.037	-0.005

The naive estimator overstates the persuasion effect by 135% of the truth. The oracle and the modular cell contrast recover the truth.

Figure 1 traces the bias of each estimator across $\lambda_{\text{fit}} \in \{0, 0.25, 0.5, 0.75, 1.0, 1.5, 2.0\}$, averaged over 25 Monte Carlo replications per value with ± 1 SE shaded bands.

The naive bias is non-monotonic: it rises sharply through $\lambda_{\text{fit}} \in [0, 0.75]$ to a peak of approximately 0.05, then attenuates over $[0.75, 2.0]$. This pattern matches the closed-form expression in Proposition 4. The numerator of the bias term is linear in λ_{fit}, but the denominator is quadratic, so the bias is maximized at $\lambda^\star_{\text{fit}} = \sigma_E / \sqrt{\text{Var}(\tilde{Q})}$, which equals approximately 0.5 for our parameter values. The oracle estimator and the modular cell contrast remain near zero throughout. These estimates suggest that a researcher who relies on realized-expression regressions cannot tell where on the bias curve they sit; what they need is a design that bypasses the bias entirely.

[6] We do not interpret the retrieval-dominant DGP as evidence about real LLM recommenders, only as a stress test of the design. The modular estimator tracks the truth across two DGPs that differ in which channel is leading; we read this as evidence that the recovery is structural rather than parameter-tuned. See Appendix E for per-category panels and additional diagnostics.

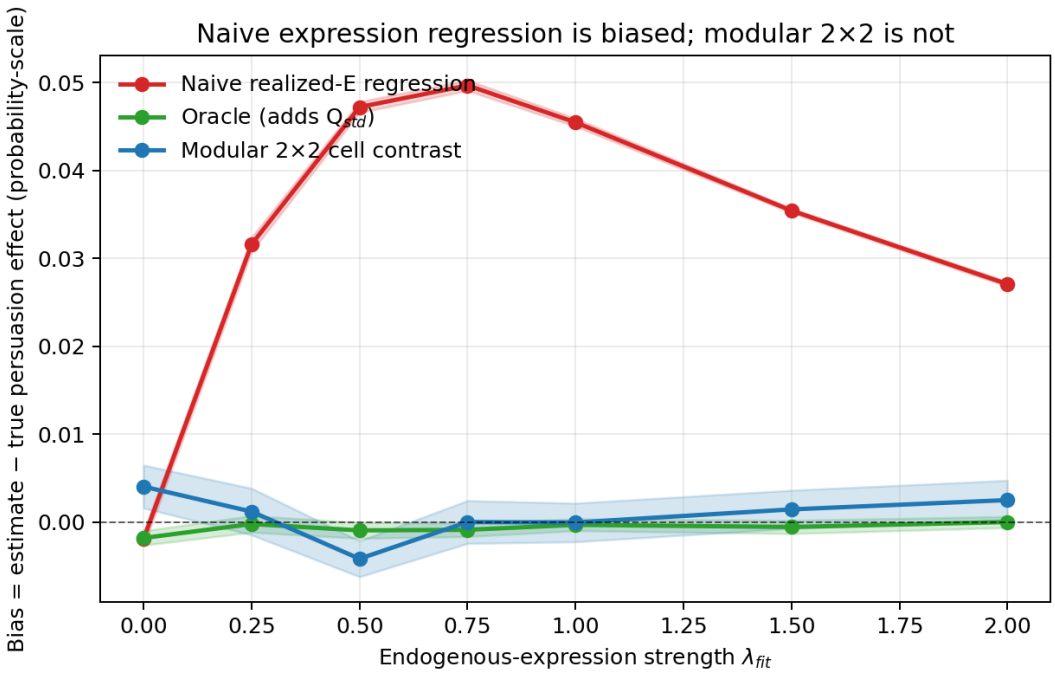

Figure 1: Bias of the naive realized-expression regression, the oracle regression, and the modular 2×2 cell contrast as a function of the expression-fit endogeneity parameter λ_{fit}. Each point is the mean over 25 Monte Carlo replications; shaded bands are ± 1 SE. Naive bias rises sharply through $\lambda_{\text{fit}} \in [0, 0.75]$ to a peak of 0.05 and then attenuates. The oracle and modular estimators stay near zero throughout.

6.5 LLM-based simulation (placeholder)

We will use a deployed LLM to generate one-shot recommendations and modular retrieval/expression outputs in three settings and to compute the corresponding empirical analogue of the structural simulation. The LLM-based simulation will calibrate the structural parameters — in particular λ_{fit} and σ_R — against the empirically observed expression-endogeneity strength of a deployed LLM recommender. This section will be completed before the conference submission deadline and is documented in detail in Appendix F.

7 Conclusion

We propose a modular 2×2 experimental design that crosses an LLM retrieval policy with an LLM expression policy, and we show that the design identifies the retrieval, persuasion, and retrieval-persuasion interaction effects over a feasible recommendation support. The framework is built on the observation that LLM-mediated product recommendation bundles two economically distinct functions: product selection and explanatory language. A standard prompt A/B test identifies the total effect of a backend policy change. The modular design separates the channels.

Our findings provide at least threefold insights with managerial relevance for firms deploying LLM shopping assistants. First, prompt A/B tests answer the wrong question when decomposition matters: they identify the total effect but leave the underlying lever bundled, and the same total effect can correspond to qualitatively different mixes of retrieval-side and expression-side lifts. Second, a firm or researcher who regresses outcomes on realized expression intensity (an analysis

that has shallow ecological validity, since the data are readily available) systematically overstates the persuasion channel, by 80–130% of the true effect at policy-relevant endogeneity levels. Third, the modular design is deployable on any platform that already issues separable retrieval and expression calls. The design affects only experimental allocation, not identification: it does not require any modification to the underlying recommendation model, only an experimenter-controlled randomization over the two policy components.

Our study is subject to several limitations that point to potential directions for future work. First, the structural simulations rely on a scalar expression intensity and on a hand-specified DGP; the LLM-based simulations described in Section 6.5 will calibrate the structural parameters against the behavior of a deployed LLM recommender. Second, all evidence in this paper is computational. A field experiment with human consumers is the natural next step, and we view the modular design as ready to deploy in such a setting. Third, the design identifies local policy effects on the feasible recommendation support; products that the baseline recommender never nominates lie outside the estimand. A potential direction for future research would be to study how to extend the support through a probabilistic shadow-logging procedure that varies the catalog grounding before the modular experiment is run.

References

Pantelis Angelopoulos, Kyusang Lee, and Sanjog Misra. Causal alignment: Augmenting language models with a/b tests. *Working paper*, 2024.

Ron Berman. Beyond the last touch: Attribution in online advertising. *Marketing Science*, 2018.

Yashar Deldjoo. Understanding biases in ChatGPT-based recommender systems. *Working paper*, 2024.

Naoki Egami, Christian J Fong, Justin Grimmer, Margaret E Roberts, and Brandon M Stewart. How to make causal inferences using texts. *Working paper*, 2018.

Or Feldman, Aishwarya Venugopal, Jann Spiess, and Amir Feder. Causal effect estimation with latent textual treatments. *Working paper*, 2026.

Dietmar Jannach. Evaluating conversational recommender systems: A landscape of research. *Artificial Intelligence Review*, 2023.

Garrett A Johnson, Randall A Lewis, and Elmar I Nubbemeyer. Ghost ads: Improving the economics of measuring online ad effectiveness. *Journal of Marketing Research*, 2017.

Hongshuang Li and P K Kannan. Attributing conversions in a multichannel online marketing environment: An empirical model and a field experiment. *Journal of Marketing Research*, 2014.

Kosuke Nakamura and Kosuke Imai. Causal representation learning with generative artificial intelligence: Application to texts as treatments. *Working paper*, 2025.

Martin Reisenbichler, Thomas Reutterer, David A Schweidel, and Daniel Dan. Frontiers: Supporting content marketing with natural language generation. *Marketing Science*, 2022.

A Proofs of Propositions

A.1 Proof of Proposition 1

By consistency, $Y_i = Y_i(J_i, E_i)$ for the realized (J_i, E_i). Under randomized assignment of Z, the joint distribution of (J, E) given $Z = z$ equals the joint distribution induced by the kernel $K_z^{1s}(\cdot \mid X, \mathcal{A})$ averaged over the population distribution of $(X, \mathcal{A})$. Therefore

$$\begin{aligned}\mathbb{E}[Y \mid Z = z] &= \mathbb{E}_{X,\mathcal{A}}\mathbb{E}_{J,E\sim K_z^{1s}(\cdot|X,\mathcal{A})}Y(J,E) \\ &= \mathbb{E}_{X,\mathcal{A}}\textstyle\sum_{j,e} K_z^{1s}(j,e \mid X,\mathcal{A})Y(j,e).\end{aligned}$$

Subtracting the $z = 0$ term from the $z = 1$ term yields Δ_{total} as defined in Equation (3). □

A.2 Proof of Proposition 2

We construct a worked counterexample. Consider two products $\{j_1, j_2\}$ and two expression states $\{e_0, e_1\}$, and write the potential outcomes as a 2×2 matrix $Y(\cdot, \cdot)$. Let the one-shot kernels be $K_0^{1s}(j, e) = 0.5 \cdot \mathbf{1}\{j = j_1, e = e_0\} + 0.5 \cdot \mathbf{1}\{j = j_2, e = e_0\}$ and $K_1^{1s}(j, e) = 0.5 \cdot \mathbf{1}\{j = j_1, e = e_1\} + 0.5 \cdot \mathbf{1}\{j = j_2, e = e_1\}$. Then the total effect is

$$\Delta_{\text{total}} = \tfrac{1}{2}[Y(j_1, e_1) - Y(j_1, e_0)] + \tfrac{1}{2}[Y(j_2, e_1) - Y(j_2, e_0)]\,.$$

Now consider two potential-outcome arrays. **Array A** has $Y(j_1, e_0) = 0.2$, $Y(j_1, e_1) = 0.4$, $Y(j_2, e_0) = 0.3$, $Y(j_2, e_1) = 0.5$. **Array B** has $Y(j_1, e_0) = 0.3$, $Y(j_1, e_1) = 0.5$, $Y(j_2, e_0) = 0.2$, $Y(j_2, e_1) = 0.4$. Both arrays yield $\Delta_{\text{total}} = 0.2$. But the implied modular cell means differ: array A has $\Delta_R = 0$ and $\Delta_P = 0.2$; array B has the same Δ_P but a non-zero $\Delta_R\times$ off-diagonal pattern because the product-expression match is different. A researcher observing only Z cannot distinguish array A from array B. □[7]

A.3 Proof of Proposition 3

Randomized assignment of (Q, R) implies $Q, R \perp \{Y(j, e) : j \in \mathcal{A}, e \in E\} \mid X, \mathcal{A}$. By consistency, $\mathbb{E}[Y \mid Q = q, R = r] = \mathbb{E}_{X,\mathcal{A}}\mathbb{E}_{J,E\sim K_{q,r}^{2s}(\cdot|X,\mathcal{A})}Y(J, E)$, which by the modular factorization in Equation (2) equals the expected potential outcome under modular policy (q, r). Taking the four cell-mean contrasts displayed in the Proposition yields the three estimands. Overlap guarantees that each cell mean is identified in the limit. □

A.4 Proof of Proposition 4

Standard omitted-variable bias for OLS. The full population regression $Y = \beta_0 + \beta_E E + \beta_Q \tilde{Q} + \nu$ is correctly specified. The naive regression omits $\tilde{Q}$, and the OLS coefficient on E satisfies

$$\operatorname{plim} \hat{\beta}_E^{\text{naive}} = \beta_E + \frac{\operatorname{Cov}(E, \tilde{Q})}{\operatorname{Var}(E)}\beta_Q.$$

With $E = e_0 + \tau R + \lambda_{\text{fit}}\tilde{Q} + \sigma_E \eta$ and $\eta, \tilde{Q}, R$ mutually independent, $\operatorname{Cov}(E, \tilde{Q}) = \lambda_{\text{fit}}\operatorname{Var}(\tilde{Q})$ and $\operatorname{Var}(E) = \lambda_{\text{fit}}^2\operatorname{Var}(\tilde{Q}) + \sigma_E^2 + \tau^2\operatorname{Var}(R)$. Substituting yields the closed form. The bias maximum follows from a first-order condition on the closed-form expression. □

[7] The counterexample illustrates the broader point that the one-shot policy realizes only the two diagonal cells (K_0^{1s}) and (K_1^{1s}) of the modular kernel; the off-diagonal hybrid cells $(K_{1,0}^{2s}, K_{0,1}^{2s})$ that distinguish Δ_R from Δ_P are not observable from one-shot data.

B Structural DGP specification

B.1 Retrieval kernel

Under the persuasion-leaning DGP, the retrieval score for consumer i and product j is

$$\text{score}_{ij}(q) = a_Q \tilde{Q}_{ij} + a_{\text{inc}}\text{incumbent}_j + a_{\text{focal}} \cdot q \cdot \text{focal_brand}_j + \sigma_R \varepsilon_{ij},$$

where $\tilde{Q}_{ij}$ is the within-category standardized fit score and $\varepsilon_{ij} \sim \mathcal{N}(0,1)$. The selected product is $J_i(q) = \arg\max_j \text{score}_{ij}(q)$. Under the retrieval-dominant DGP, the a_{focal} term is replaced by an increase in the Q-weighting under $q = 1$: $\text{score}_{ij}(q) = (a_Q + \delta_{\text{fit}} \cdot q)\tilde{Q}_{ij} + a_{\text{inc}}\text{incumbent}_j + \sigma_R \varepsilon_{ij}$.

B.2 Expression kernel

One-shot: $E_i(z) = e_0 + \tau_{\text{backend}} z + \lambda_{\text{fit}} \tilde{Q}_{i,J_i(z)} + \lambda_{\text{inc}}\text{incumbent}_{J_i(z)} + \sigma_E \eta_i$. Modular: $E_i(q,r) = e_0 + \tau_R r + \lambda_{\text{fit}} \tilde{Q}_{i,J_i(q)} + \lambda_{\text{inc}}\text{incumbent}_{J_i(q)} + \sigma_E \eta_i$. The bundle asymmetry — z enters both the retrieval and the expression equations in the one-shot DGP, while q and r enter only their own equations in the modular DGP — is the structural feature that makes Proposition 2 bite.

B.3 Outcome equation

$Y_i \sim \text{Bernoulli}(\sigma(U_i))$ with $U_i = \beta_0 + \beta_Q \tilde{Q}_{i,J_i} + \beta_E E_i + \beta_{\text{inc}}\text{incumbent}_{J_i} + \beta_{\text{incE}}\text{incumbent}_{J_i} E_i$. We do not include β_{QE} in the main run; setting it nonzero produces a non-zero true interaction and is documented in Appendix E.

B.4 Parameter values

Main DGP		Robustness DGP	
$a_Q = 1.0$	$a_{\text{inc}} = 0.20$	$a_Q = 1.0$	$a_{\text{inc}} = 0.20$
$a_{\text{focal}} = 0.50$	$\sigma_R = 0.35$	$a_{\text{focal}} = 0$	$\delta_{\text{fit}} = 0.80$
$e_0 = 0$	$\tau_{\text{backend}} = 0.50$	$e_0 = 0$	$\tau_{\text{backend}} = 0.50$
$\tau_R = 0.50$	$\lambda_{\text{fit}} = 1.0$	$\tau_R = 0.50$	$\lambda_{\text{fit}} = 1.0$
$\lambda_{\text{inc}} = 0.20$	$\sigma_E = 0.50$	$\lambda_{\text{inc}} = 0.20$	$\sigma_E = 0.50$
$\beta_0 = -0.80$	$\beta_Q = 0.70$	$\beta_0 = -0.80$	$\beta_Q = 0.70$
$\beta_E = 0.50$	$\beta_{\text{inc}} = 0.25$	$\beta_E = 0.35$	$\beta_{\text{inc}} = 0.25$
$\beta_{\text{incE}} = -0.15$	$\beta_{QE} = 0$	$\beta_{\text{incE}} = -0.15$	$\beta_{QE} = 0$

B.5 Seed handling

We use the master seed 20260513 with `numpy.random.SeedSequence.spawn(5)` to derive five independent streams for retrieval, one-shot expression, one-shot outcome, modular expression, and modular outcome. The λ_{fit} sweep further spawns 25 Monte Carlo seeds per λ value for the expression and outcome streams; the retrieval realization is shared across the sweep.

C Catalog and consumer profile generation

C.1 Catalog construction and validation

For each of three categories (phone chargers, headphones, laptops) we construct a catalog of 8–10 products with brand, brand status (incumbent or entrant), price, quality score, attributes, review-summary text, use-case fit vectors, focal-brand indicator, sponsored-product indicator, and weakness.

Catalogs are validated to ensure: (i) no single product dominates on more than two of six use cases; (ii) prices span at least a 3× ratio per category; (iii) at least four distinct best-fit products by use case. Generation is deterministic; the catalog files are checked into the repository.

C.2 Consumer-profile generator

We draw 1,000 consumer profiles per category using a master seed of 42. Each profile has: a lognormal budget; uniform price and quality sensitivities on $[0.2, 1.0]$; Beta-distributed brand familiarity with parameters $(6, 3)$ for incumbents and $(2, 6)$ for entrants, plus a tech-savviness boost of $+0.15$ for entrant familiarity; a use-case drawn from a category-specific categorical distribution; and three behavioral parameters (persuasion susceptibility, comparison preference, trust in AI) drawn from uniform $[0, 1]$.

C.3 Latent fit score

The latent fit score is $Q_{ij} = w_p \cdot \text{price_sensitivity}_i \cdot \text{price_fit}_{ij} + w_q \cdot \text{quality_sensitivity}_i \cdot \text{quality_score}_j/100 + w_u \cdot \text{use_case_fit}_{ij} + w_b \cdot \text{brand_familiarity}_{ij}$, with category-specific weights w_p, w_q, w_u, w_b that sum to one. The score is within-category standardized before being used in retrieval and outcome equations.

D Estimator details

D.1 Cell-mean contrasts and analytical standard errors

For each modular cell $(q, r) \in \{0, 1\}^2$ we compute the cell mean $\bar{Y}_{q,r}$ and the cell variance $s^2_{q,r}$. The standard error of a two-cell contrast is $\sqrt{s^2_{q_1,r_1}/n_{q_1,r_1} + s^2_{q_2,r_2}/n_{q_2,r_2}}$. The standard error of the four-cell interaction contrast is $\sqrt{\sum_{(q,r)} s^2_{q,r}/n_{q,r}}$.

D.2 Naive and oracle regressions

Both regressions are OLS linear probability models with HC1 (Stata-equivalent) robust standard errors. The naive regression is $Y \sim E + \text{incumbent} + \text{focal_brand} + \text{category FE}$; the oracle regression adds $\tilde{Q}_J$ as a control. The coefficient on E is rescaled to the probability scale of a $R = 0 \to R = 1$ policy change by multiplying by the within-modular $(Q = 0)$ mean shift $\bar{E}_{Q=0,R=1} - \bar{E}_{Q=0,R=0}$.

D.3 Monte Carlo replication

The headline figure (Figure 1) uses $M = 25$ Monte Carlo replications per λ_{fit} value. The retrieval realization is shared across reps; only the expression noise and the outcome Bernoulli draws are resampled. Standard errors of the mean bias across reps are computed as the empirical standard deviation divided by $\sqrt{M}$.

E Additional tables and figures

E.1 Per-category modular decomposition

The per-category breakdown of Table 2 is reproduced from the simulation output. With $N = 1{,}000$ per category and four modular cells, per-category standard errors are approximately 0.022 for cell-mean contrasts, leaving most per-category t-statistics at the edge of detectability. We pool across cate-

gories for the main paper. Per-category panels are saved in `results/tables/decomposition_mvp.csv` and `results/tables/robustness_decomposition.csv`.

E.2 Lambda-sweep raw values

The aggregated λ_{fit} sweep used in Figure 1 is saved in `results/tables/lambda_sweep_aggregated.csv`. The 175 per-replication rows underlying the figure are saved in `results/tables/lambda_sweep_raw.csv`.

E.3 Robustness DGP detail tables

The full robustness DGP output (per-category and pooled decompositions; naive/oracle/modular estimates) is saved in `results/tables/robustness_decomposition.csv`, `results/tables/robustness_naive_vs_or` and `results/tables/robustness_one_shot_total_effect.csv`.

F LLM-based simulation protocol

This appendix will be completed concurrently with Section 6.5. The current plan is documented in `plans/llm_simulation_plan.md` in the project repository. The protocol covers: (i) model and quantization choice; (ii) prompt templates and JSON schema for retrieval-stage outputs; (iii) expression-stage prompts; (iv) calibration of λ_{fit} and σ_R against the empirical behavior of the deployed LLM; (v) reproducibility notes.